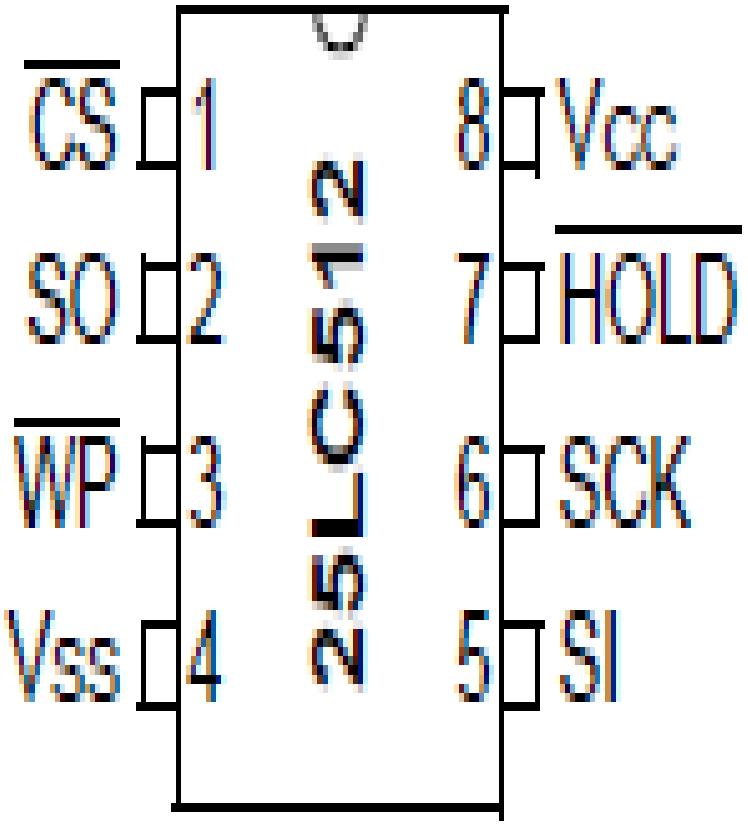
Implementing SPI communication algorithms for EEPROM **25LC512**

Referring Data Sheet

Features

- **512 Kbit (65536 bytes) SPI Bus Serial EEPROM**
- **VCC Range 2.5 – 5.5 v**
- 20 MHz max. Clock Speed
- Byte and Page-level Write Operations:
  - 128-byte page
  - 5 ms max.
- Self-Timed Erase and Write cycles:
  - Page Erase (5 ms, typical)
  - Sector Erase (10 ms/sector, typical)
  - Bulk Erase (10 ms, typical)
- Sector Write Protection (16K byte/sector):
  - Protect none, 1/4, 1/2 or all of array
- Built-In Write Protection:
  - Power-on/off data protection circuitry
  - Write enable latch
  - Write-protect pin
- Supports SPI Modes 0(0,0) & 3(1,1)
- Supports Serial Data Transfer Order MSB first

Pin Diagram



Pin Function Table

Name	Function
$\overline{CS}$	Chip Select Input
SO	Serial Data Output
$\overline{WP}$	Write-Protect
Vss	Ground
SI	Serial Data Input
SCK	Serial Clock Input
$\overline{HOLD}$	Hold Input
Vcc	Supply Voltage

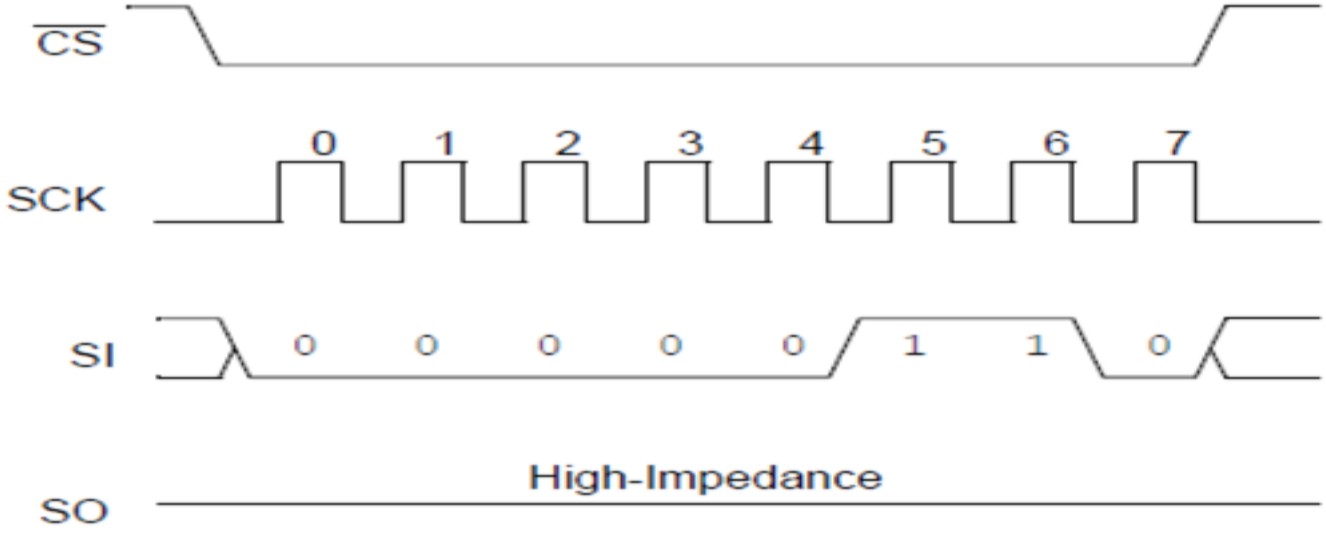
Instruction Set

n Name	Instruction Format	Description
D	0000 0011	Read data from memory array beginning at selected
TE	0000 0010	Write data to memory array beginning at selected ad
N	0000 0110	Set the write enable latch (enable write operations)
I	0000 0100	Reset the write enable latch (disable write operation
R	0000 0101	Read STATUS register
R	0000 0001	Write STATUS register
	0100 0010	Page Erase – erase one page in memory array
	1101 1000	Sector Erase – erase one sector in memory array
	1100 0111	Chip Erase – erase all sectors in memory array
D	1010 1011	Release from Deep power-down and read electronic
D	1011 1001	Deep Power-Down mode

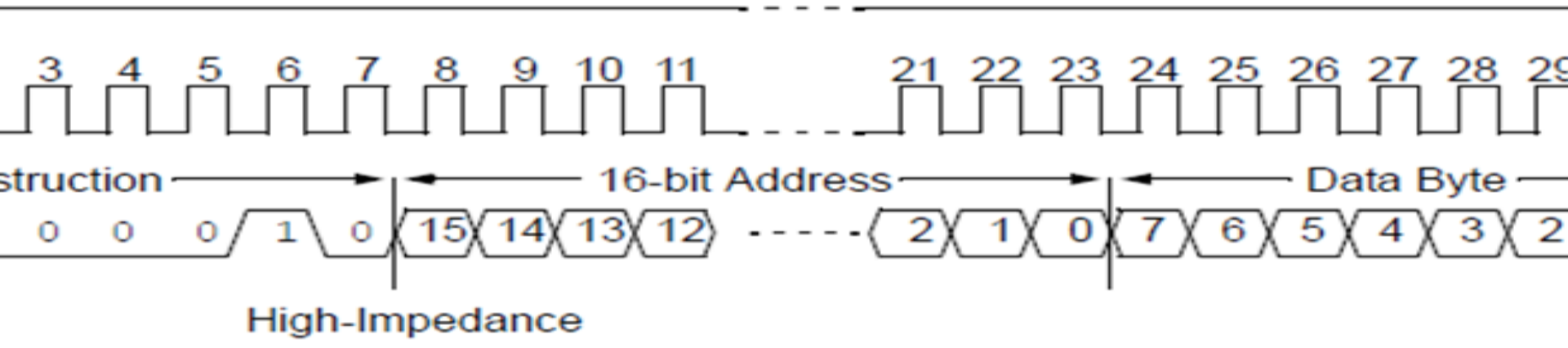
Write Sequence

- Prior to any attempt to write data to the 25LC512, the write enable latch must be set by issuing the WREN instruction.

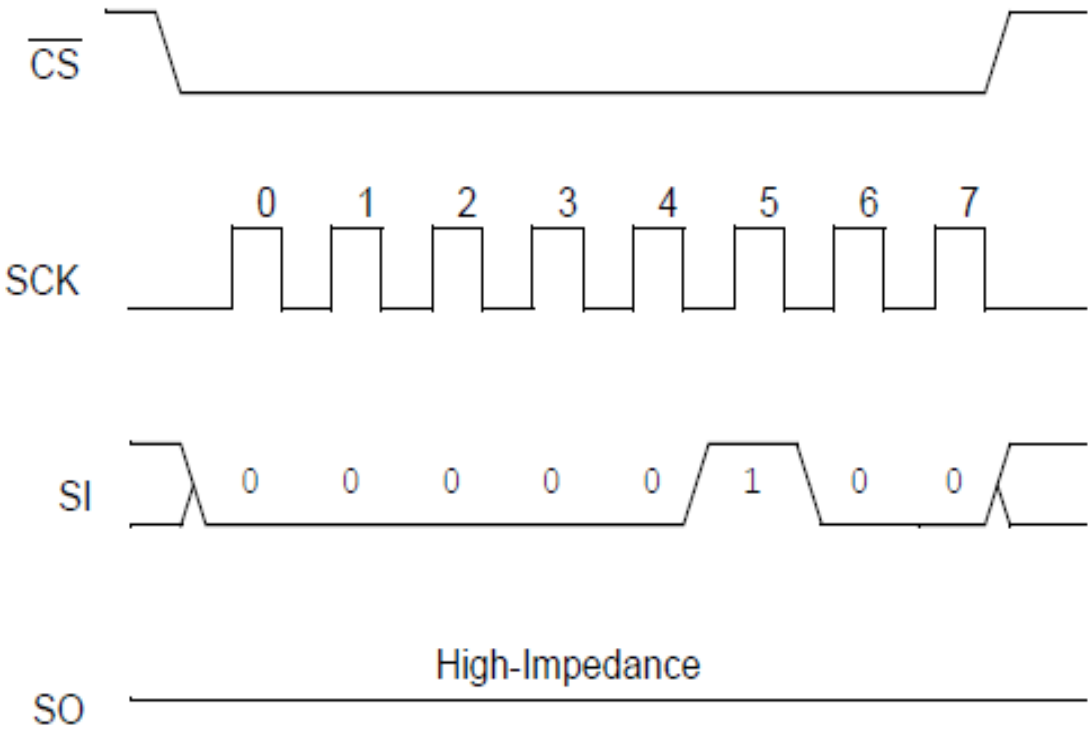
WRITE ENABLE SEQUENCE (WREN)



BYTE WRITE SEQUENCE



WRITE DISABLE SEQUENCE (WRDI)



READ SEQUENCE

